

1. For each of the following sequences (U_n) whose n -th term is given below, calculate the three first terms and determine expressions of U_{2n} and U_{n+1} .

- $U_n = \frac{1^2+2^2+3^2+\dots+n^2}{n^3}$, $U_n = \frac{1}{(n-2)(n+1)}$, $U_n = \frac{1.3.5\dots(2n-1)}{2.4\dots(2n)}$, $U_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n}$.

2. Study the monotony of the following sequences:

$$U_n = \frac{2^n}{n!}, \quad U_n = n + (-1)^n, \quad U_n = 2n + (-1)^n, \quad U_n = n^2 - 2(-1)^n$$

3. By using (ε, N) definition of the limit, show that:

- $\lim_{n \rightarrow +\infty} na^n = 0$ ($|a| < 1$), $\lim_{n \rightarrow +\infty} \sqrt[n]{a} = 1$ ($a > 1$), $\lim_{n \rightarrow +\infty} \sqrt[n]{n} = 1$.
- $\lim_{n \rightarrow +\infty} \frac{\ln(n+1)}{\ln n} = 1$, $\lim_{n \rightarrow +\infty} \frac{\ln(1+e^n)}{2n} = \frac{1}{2}$, $\lim_{n \rightarrow +\infty} 3^{2n+1} = +\infty$.

4. Determine the limit of each of the following sequences.

- $U_n = \frac{\sin(n)}{n}$, $U_n = \frac{n^n}{n!}$, $U_n = \frac{E(n^2 \sin(\frac{1}{n}))}{2n+1}$, $U_n = \frac{1+a+a^2+\dots+a^n}{1+b+b^2+\dots+b^n}$ ($|a| < 1, |b| < 1$),
- $U_n = \frac{3^n - 7^n}{3^n + 7^n}$, $U_n = \sum_{k=1}^n \frac{1}{\sqrt{n^2+1}}$, $U_n = \sum_{k=0}^n \frac{k!}{(n+2)!}$
- $U_n = \sum_{k=1}^n \frac{1}{\sqrt{k}}$, *Hint.* See that $\frac{1}{2\sqrt{k}} \geq \sqrt{k+1} - \sqrt{k}$.

5. Let $\{u_n\}_{n \in \mathbb{N}}$ be a numerical sequence. Show that:

- (a) $\{u_{2n}\}_{n \in \mathbb{N}}$ and $\{u_{2n+1}\}_{n \in \mathbb{N}}$ are both convergent to the same limit ℓ **if and only if** $\{u_n\}_{n \in \mathbb{N}}$ is convergent to ℓ .
- (b) If $\{u_n\}_{n \in \mathbb{N}}$ is increasing $\{u_{2n}\}_{n \in \mathbb{N}}$ is convergent, then $\{u_n\}_{n \in \mathbb{N}}$ is convergent.

6. Determine $\overline{\lim}_{n \rightarrow \infty} a_n$ and $\underline{\lim}_{n \rightarrow \infty} a_n$, where

- (1) $a_n = n^{(-1)^n n}$,
 (2) $a_n = 1 + n \sin \frac{n\pi}{2}$,
 (3) $a_n = \left(2 \cos \frac{2n\pi}{3}\right)^n$.

7. Let $(U_n)_{n \in \mathbb{N}}$ be a numerical sequence of some kind. Do we have the equivalence

$$(U_n)_{n \in \mathbb{N}} \text{ is convergent} \Leftrightarrow \lim_{n \rightarrow +\infty} (U_{n+1} - U_n) = 0 ?$$

8. Use the criterion of monotonic sequence to study the nature of the sequences of n -th term:

$$U_n = \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots (2n)}, \quad U_n = \sum_{k=1}^n \frac{1}{k^2}.$$

9. Consider the sequence defined recursively by

$$x_1 = \sqrt{2}, \quad x_n = \sqrt{2 + x_{n-1}}.$$

- (a) Show, by induction, that $x_n < 2$ for all n .
- (b) Show, by induction, that $x_n < x_{n+1}$ for all n .
- (c) Find the limit of $\{x_n\}$.

10. Study the nature of the sequences of the n -th term

$$\left\{ \begin{array}{l} u_0 = \alpha \in \mathbb{R}, \\ u_{n+1} = u_n + 1, \quad \forall n \in \mathbb{N}, \end{array} \right\} \quad \left\{ \begin{array}{l} u_0 = \alpha \in \mathbb{R}, \\ u_{n+1} = \frac{u_n}{2} + 1, \quad \forall n \in \mathbb{N}, \end{array} \right\} \quad \left\{ \begin{array}{l} u_0 = 1, \\ u_{n+1} = \sqrt{u_n + 2}, \quad \forall n \in \mathbb{N}. \end{array} \right\}$$

11. — Show that in each of the following cases the sequences (u_n) and (v_n) are adjacent.

- $u_n = 2^n \sin \frac{\theta}{2^n}$ and $v_n = 2^n \tan \frac{\theta}{2^n}$, for $n \geq 1$, $\theta \in]0, \frac{\pi}{2}[$,
- $u_n = \left(1 + \frac{1}{n}\right)^n$, $v_n = \left(1 + \frac{1}{n}\right)^{n+1}$, for $n \geq 1$.

12. For $k \in]0, 1[$, let $\{u_n\}_{n \in \mathbb{N}}$ be a sequence satisfying

$$|u_{n+1} - u_n| \leq k |u_n - u_{n-1}|, \quad \forall n \geq 1.$$

Such a sequence is called contractant.

(a) Show that

$$|u_{n+1} - u_n| \leq k^n |u_1 - u_0|, \quad \forall n \geq 1.$$

(b) Show that $\{u_n\}_{n \in \mathbb{N}}$ is convergent.

13. By using Cauchy's criterion show that the sequences of n -th term are convergent or divergent.

$$U_n = \sum_{k=1}^{k=n} \frac{\sin(k)}{k^2}, \quad U_n = \sum_{k=1}^{k=n} \frac{\sin(k)}{3^k}, \quad U_n = \sum_{k=1}^{k=n} \frac{1}{\sqrt{k}}.$$

14. Show that the subsequences (U_{2n}) and (U_{2n+1}) are adjacent, then conclude for (U_n) , where

$$U_n = \sum_{k=1}^{k=n} \frac{(-1)^k}{k}.$$

15. Let $(U_n)_{n \geq 1}$ be a sequence and let (V_n) be defined by $V_n = \frac{U_1 + U_2 + \cdots + U_n}{n}$, $n \geq 1$. Show that if $\lim_{n \rightarrow +\infty} U_n = l$, then $\lim_{n \rightarrow +\infty} V_n = l$.

16. Let $\{u_n\}$ be a real sequence, and $\ell \in \mathbb{R}$.

- (a) If $\lim_{n \rightarrow \infty} (u_n - u_{n-1}) = \ell$, then $\lim_{n \rightarrow \infty} \frac{u_n}{n} = \ell$.
- (b) If $\{u_n\}$ has only positive terms and $\lim_{n \rightarrow \infty} u_n = \ell$, then $\lim_{n \rightarrow \infty} \sqrt[n]{u_1 u_2 \cdots u_n} = \ell$.

— Find the limit, as $n \rightarrow +\infty$, of

$$\frac{\ln(2) + 2^{-1} \ln(3) + \cdots + n^{-1} \ln(n+1)}{\ln(n)}.$$

17. Let $\{u_n\}_{n \in \mathbb{N}}$ be a sequence of only positive terms such that $\{\sqrt[n]{u_n}\}_{n \in \mathbb{N}}$ converge to ℓ .

- (a) Show if $\ell < 1$, then $\lim_{n \rightarrow +\infty} u_n = 0$.
- (b) Show that if $\ell > 1$, then $\lim_{n \rightarrow +\infty} u_n = +\infty$.
- (c) Show that the case $\ell = 1$ is not conclusive.

— Find the limit of $2^{n^2}/(n!)^n$ as $n \rightarrow +\infty$.

18. Let $\{u_n\}_{n \in \mathbb{N}}$ be a sequence of only positive terms.

- (a) Show that

$$\left\{ \frac{u_{n+1}}{u_n} \right\}_{n \in \mathbb{N}} \text{ converge to } \ell \implies \{\sqrt[n]{u_n}\}_{n \in \mathbb{N}} \text{ also converges to } \ell.$$

- (b) Show, by means of counterexample, that the inverse is not true.

— Find the limit, as $n \rightarrow +\infty$, of the following:

$$\frac{n}{\sqrt[n]{n!}}, \quad \frac{1}{n} \sqrt[n]{n(n+1) \dots (n+n)}, \quad \frac{1}{n} \sqrt[n]{1.3 \dots (2n-1)}, \quad \frac{1}{n^2} \sqrt[n]{\frac{(3n)!}{n!}}.$$

19. Let $\{u_n\}_{n \in \mathbb{N}}$ be a sequence of only positive terms such that $\left\{ \frac{u_{n+1}}{u_n} \right\}_{n \in \mathbb{N}}$ converge to ℓ .

- (a) Show if $\ell < 1$, then $\lim_{n \rightarrow +\infty} u_n = 0$.
- (b) Show that if $\ell > 1$, then $\lim_{n \rightarrow +\infty} u_n = +\infty$.
- (c) Show that the case $\ell = 1$ is not conclusive.

— Find the limit of $\frac{(an)!}{n^{an}}$, $\frac{a^n}{n!}$ and $n^a \cos^n(\theta)$ as $n \rightarrow +\infty$ for $a > 0$ and $\theta \in]0, \pi/2[$.

20. Let (x_n) and (y_n) be the sequences defined for $n \in \mathbb{N}$ by

$$x_n = \sqrt{2n} - \sqrt{2n-1} \text{ and } y_n = \sqrt{n + (-1)^{n+1}} - \sqrt{n}.$$

Set

$$\Delta = \{x_n : n \in \mathbb{N}\} \text{ and } \Omega = \{y_n : n \in \mathbb{N}\}.$$

- (a) Show that Δ is bounded and determine $\max \Delta$, $\sup \Delta$, $\inf \Delta$ and $\min \Delta$ if they exist.
- (b) Find the relationship between Δ and Ω , then conclude for $\max \Omega$, $\sup \Omega$, $\inf \Omega$ and $\min \Omega$.

21. Prove that the sequence (U_n) defined by:

$$\begin{cases} U_0 = 0, \\ U_n = \frac{1}{2} \sin(U_{n-1}) + \frac{1}{3} \cos(U_{n-1}) \text{ for all } n \in \mathbb{N} \end{cases}$$

is convergent.